

An Efficient Iterative Detection and Decoding Receiver for Polar-Coded Massive MIMO

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Abstract—This paper presents an efficient iterative detection and decoding (IDD) receiver for polar-coded massive MIMO systems. The EP with approximation (EPA) algorithm is devised to relieve the computational burden of the detector. A modified belief propagation flip (mBPF) polar decoder is proposed to effectively improve the decoding performance with acceptable complexity cost. The messages passed between the detector and decoder are carefully designed to further improve the bit-error-rate (BER) performance. Numerical results are presented to demonstrate that the proposed IDD scheme can outperform both the separate detection and decoding (SDD) receiver with EP detector and successive cancellation list (SCL) decoder, and the state-of-the-art (SOA) double EP (DEP) IDD receiver with BP decoder. The corresponding complexity analysis and hardware architecture are also provided in this paper.

Index Terms—Massive MIMO, expectation propagation, iterative detection and decoding, belief propagation, critical set

I. INTRODUCTION

Massive multiple-input multiple-output (MIMO) is a key technology to meet the demands of 5G and beyond for its capability to attain high data rates and link reliability [1]. It is usually concatenated with powerful channel codes to achieve near-capacity performance. For coded MIMO, iterative detection and decoding (IDD) receiver has been demonstrated to provide better error rate performance than separated detection and decoding (SDD) schemes. However, the high complexity hinders the practical implementation of IDD receivers.

There are many challenges for designing an implementation friendly IDD receiver, especially for polar-coded MIMO systems. First, with the increase of system size, it is difficult to balance the complexity and performance of the MIMO detector. Recently, the near-optimal expectation propagation detector (EPD) [2] has been considered for IDD to improve detection accuracy. The double EP (DEP) [3] algorithm designed for low-density parity-check (LDPC)-coded MIMO is shown to achieve promising performance, which is considered the current state-of-the-art (SOA) EP-based IDD scheme. Nevertheless, it involves an inevitable matrix inversion of cubic complexity in each detection iteration. The efficient EP with approximation (EPA) detector [4] is also applied with IDD [5], but limited to LDPC-coded scenarios.

On the other hand, the research concerning IDD for polar-coded systems is mainly limited to the decoding module. As the preferred successive cancellation list (SCL) decoder

only produces hard output, a soft-output list decoder (SOL) is proposed in [6] to provide soft messages required in IDD for polar codes. Nevertheless, the SOL decoder still suffers from intolerable latency and complexity. The belief propagation (BP) [7] decoder is inherently parallel and easier to implement, and can be naturally adopted for IDD to support soft output with lower latency, but with a performance far from satisfactory. A linear programming based initialization is proposed in [8] to enhance the performance of DEP-BP, which brings unaffordable extra computational burden for large systems. The BP flip (BPF) decoding proposed in [9] offers an option to improve the performance of BP through additional decoding attempts with a few bits flipped in critical set (CS) [10], but its capability in IDD has not yet been investigated.

In this paper, an efficient EPA-based IDD receiver is first proposed for polar-coded massive MIMO systems. Our main contributions are as follows. (1) To relieve the computational burden while maintaining good performance, EPA is adopted for detection with subtle design of the messaging between detector and decoder. (2) To further reduce the decoding complexity and latency, we propose a modified BP flip (mBPF) method to improve the decoding performance in the IDD scheme, which is shown to resolve the error-floor issue of BP with acceptable complexity. (3) An efficient hardware architecture is presented for the above EPA-mBPF IDD receiver.

II. PRELIMINARIES

A. System Model

Consider an uplink polar-coded MIMO system equipped with N_r receiving antennas serving N_t single-antenna users. The transmitted information bit vector $\mathbf{a}$ is encoded into codeword vector $\mathbf{b}$ by the (N, K) polar code. $\mathbf{b}$ is interleaved and mapped into the M -QAM constellation Θ , which is then split into sets of vectors $\mathbf{x} \in \Theta^{N_t \times 1}$ and sent over a MIMO channel $\mathbf{H} \in \mathbb{C}^{N_r \times N_t}$. The received signal $\mathbf{y} \in \mathbb{C}^{N_r \times 1}$ is obtained by

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}, \quad (1)$$

where $\mathbf{n} \sim \mathcal{CN}(0, \sigma_n^2 \mathbf{I}_{N_r})$ is the additive white Gaussian noise (AWGN) with zero mean and variance σ_n^2 . Here, we consider the practical correlated MIMO channels by the Kronecker model [12]. The correlation coefficients on the receiver and transmitter sides are denoted by ζ_R and ζ_T , respectively. It is

supposed that perfect channel state information (CSI) is known at the receiver. By real-value decomposition, the posterior distribution of $\mathbf{x}$ is modeled as

$$p(\mathbf{x}|\mathbf{y}, \mathbf{H}) \propto \mathcal{N}(\mathbf{y} : \mathbf{H}\mathbf{x}, \sigma_n^2 \mathbf{I}_{2N_r}) \prod_{i=1}^{2N_t} p_r(x_i), \quad (2)$$

where $p_r(x_k)$ is the prior PDF of x_k which is usually assumed equiprobable as $p_r(x_k) = \frac{1}{\Theta} \mathbb{I}_{x_i \in \Theta}$.

B. EP-based Detector and IDD Schemes

The motivation of EP [2] is to approximate the intractable Eq. (2) by the unnormalized Gaussian distribution:

$$q(\mathbf{x}) \propto \mathcal{N}(\mathbf{y} : \mathbf{H}\mathbf{x}, \sigma_n^2 \mathbf{I}_{2N_r}) \prod_{i=1}^{2N_t} \exp(-\frac{1}{2} \Lambda_i x_i^2 + \gamma_i x_i). \quad (3)$$

For each EP iteration, the mean $\boldsymbol{\mu}$ and variance $\boldsymbol{\Sigma}$ of $q(\mathbf{x})$ are first computed as:

$$\boldsymbol{\Sigma} = (\mathbf{A} + \boldsymbol{\Lambda})^{-1}, \quad \boldsymbol{\mu} = \boldsymbol{\Sigma}(\tilde{\mathbf{b}} + \boldsymbol{\gamma}), \quad (4)$$

where $\boldsymbol{\gamma} = [\gamma_1, \dots, \gamma_{2N_t}]$, $\boldsymbol{\Lambda} = \text{diag}([\Lambda_1, \Lambda_2, \dots, \Lambda_{2N_t}])$, $\mathbf{A} = \sigma_n^{-2} \mathbf{H}^T \mathbf{H}$, and $\tilde{\mathbf{b}} = \sigma_n^{-2} \mathbf{H}^T \mathbf{y}$. Let $\tilde{p}(x_i) = \exp(-\frac{1}{2} \Lambda_i x_i^2 + \gamma_i x_i)$, the cavity distribution is then formed by:

$$q^{\setminus i}(x_i) = q(x_i) / \tilde{p}(x_i) \propto \mathcal{N}(t_i, h_i^2). \quad (5)$$

The Gaussian parameters (Λ_i, γ_i) are then updated through the following moment matching (MM) rule:

$$\tilde{p}^{new}(x_i) = \text{proj}_{\mathcal{G}} \left(q^{\setminus i}(x_i) p_r(x_i) \right) / q^{\setminus i}(x_i), \quad (6)$$

where $\mathcal{G}$ denotes the Gaussian family.

Each update of Eq. (4) has a computational cost of $\mathcal{O}(N_t^3)$, which hinders the efficient implementation of EPD. To reduce the complexity, EPA [4] reformulates the MM equation with approximation to eliminate the iterative matrix inversion of Eq. (4). Specifically, let $t_i = \rho_i / v_i$ and $h_i^2 = 1 / v_i$, and denote the mean and variance of $\text{proj}_{\mathcal{G}}(q^{\setminus i}(x_i) p_r(x_i))$ by (η_i, χ_i) , the MM procedure can be reformed as [4]:

$$\mathbf{m} = \mathbf{y} - \mathbf{H}\boldsymbol{\eta}, \quad \boldsymbol{\rho} = \mathbf{H}^T \mathbf{m} / \sigma_n^2 + \mathbf{v}\boldsymbol{\eta}. \quad (7)$$

By exploiting the channel hardening phenomenon, the above v_i can be further approximated by $v_i \approx \{\mathbf{H}^T \mathbf{H}\}_{ii} / \sigma_n^2 = A_{ii}$, which is constant. Therefore, in EPA, the computation of Eq. (4) is only required once to initialize ρ_i . The rest of iterations only involve the updates of (η_i, χ_i) and Eq. (7).

The current SOA EP-based IDD scheme is DEP [3], which follows the following turbo procedure. For each outer loop, the cavity Eq. (5) forms the extrinsic messages of EPD, which are converted to bit log-likelihood ratio (LLR) and fed into the decoder. After decoding, the extrinsic LLR from the decoder is converted to symbol probabilities to replace $p_r(x_i)$. The MM Eq. (6) is then carried out with the renewed $p_r(x_i)$ to update the prior parameters (Λ_i, γ_i) , which is then used to restart the EPD inner iterations.

C. Polar BP Decoding

In polar BP decoder, messages are propagated over the generator matrix [11] $\mathbf{G}$ -based factor graph. Right-to-left messages $L_{i,k}$ and left-to-right messages $R_{i,k}$ are updated in each iteration, where i denotes row indices and k denotes stage indices. $R_{i,0}$ is initialized to $+\infty$ or 0 depending on whether the i -th bit is frozen or not. The corresponding extrinsic LLRs received from the detector are assigned to $L_{i,n}$, where $n = \log_2 N$. The matrix $\mathbf{L}$ and $\mathbf{R}$ are updated as follows in a processing element (PE).

$$\begin{cases} L_{i,k} = g(L_{i,k+1}, L_{i+2^k,k+1} + R_{i+2^k,k}), \\ L_{i+2^k,k} = g(L_{i,k+1}, R_{i,k}) + L_{i+2^k,k+1}, \\ R_{i,k+1} = g(R_{i,k}, L_{i+2^k,k+1} + R_{i+2^k,k}), \\ R_{i+2^k,k+1} = g(R_{i,k}, L_{i,k+1}) + R_{i+2^k,k}, \end{cases} \quad (8)$$

$$g(x, y) = \beta_c \cdot \min\{|x|, |y|\} \cdot \text{sgn}(x) \cdot \text{sgn}(y), \quad (9)$$

where β_c is a scaled factor to reduce the approximation error of min-sum algorithm. After the specified maximum number of iterations I_{max} is reached, we can get each estimated bit $\hat{a}_i$ through the hard decisions of $(L_{i,0} + R_{i,0})$ and $\hat{b}_i$ through the hard decisions of $(L_{i,n} + R_{i,n})$.

III. THE PROPOSED IDD SCHEME WITH POLAR CODES

A. EPA-based IDD Receiver

To construct the EPA-based IDD summarized in Alg. 1, the extrinsic messages passed from and into the detector should be carefully designed. First, since (t_i, h_i^2) are not explicitly calculated in EPA, with $t_i = \rho_i / v_i$, we reformulate Eq. (7) as

$$m_i = b_i - A_{ii}\eta_i, \quad t_i = m_i / v_i + \eta_i, \quad (10)$$

from which we explicitly obtain the extrinsic mean $t_i^E = t_i$. Moreover, the extrinsic variance h_i^{2E} could be estimated with $h_i^{2E} = 1 / v_i = 1 / A_{ii}$. The extrinsic message distribution of EPA is then approximated by $q_E(x_i) = \mathcal{N}(x_i; t_i, 1 / A_{ii})$ [5]. The output extrinsic bit-level LLRs from EPA detector is:

$$LLR(b_{k,i}) = \ln \frac{\sum_{U_k^0} \mathcal{N}(x_i; t_i, 1 / A_{ii})}{\sum_{U_k^1} \mathcal{N}(x_i; t_i, 1 / A_{ii})}, \quad (11)$$

where $b_{k,i}$ is the k -th bit of demapped x_i , U_k^b denotes the subset of Θ whose k -th bit is b .

For the t -th IDD outer loop, after decoding, the decoder output bit LLR should be converted to extrinsic symbol probabilities $p_r^{[t+1]}(x_i)$. The MM in Eq. (6) is then carried out to renew (Λ_i, γ_i) to restart the detector. For EPA IDD, the following scheme is specifically proposed. After the Gaussian projection, $(\eta_i^{[t+1,1]}, \chi_i^{[t+1,1]})$ is obtained, and the second moment is matched by $1 / \chi_i^{[t+1,1]} - 1 / h_i^{2[t+1,1]}$. If $\chi_i^{[t+1,1]} < h_i^{2[t+1,1]}$, we incorporate the MM results with message damping and update the prior parameters as

$$\Lambda_i^{[t+1,1]} = \beta \left(\frac{1}{\chi_i^{[t+1,1]}} - \frac{1}{h_i^{2[t+1,1]}} \right) + (1 - \beta) \Lambda_i^{[t,1]}, \quad (12a)$$

Algorithm 1: The proposed EPA-IDD scheme

Input : $\mathbf{A}, \tilde{\mathbf{b}}, \mathbf{v} = \text{diag}(\mathbf{A})$, $p_r^{[0]}(x_i) = \frac{1}{|\Theta|} \mathbb{I}_{x_i \in \Theta}$,
 $\varphi = 10^{-3}$, $\Lambda_i^{[0,1]} = 10^{-3}$, $\gamma_i^{[0,1]} = 0$.

Output: **a.**

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1 for  $t = 1, 2, \dots, T_{max}$  do
2   Calculate  $\Sigma^{[t,1]}$  and  $\mu^{[t,1]}$  by Eq.(4);
3   for  $i = 1, 2, \dots, N_t$  do
4      $h_i^{2[t,1]} = \Sigma_{ii}^{[t,1]} / (1 - \Sigma_{ii}^{[t,1]} \Lambda_i^{[t,1]})$ ,
5      $t_i^{[t,1]} = h_i^{2[t,1]} (\mu_i^{[t,1]} / \Sigma_{ii}^{[t,1]} - \gamma_i^{[t,1]})$ ;
6   Run EPA for  $S$  iterations.
7    $h^{2[t,S+1]} = 1/\mathbf{v}$ ;
8   Output  $q_E^{[t,S+1]}(x_i)$  and compute LLRs with Eq.(11) .
9   Run the mBPF decoder to output  $p_r^{[t+1]}(x_i)$ .
10  MM at the outer loop:
11   $\hat{p}^{[t+1,1]}(x_i) = q_E^{[t,S+1]}(x_i) p_r^{[t]}(x_i)$ ;
12   $\eta_i^{[t+1,1]} = \mathbb{E}[\hat{p}^{[t+1,1]}(x_i)]$ ,  $\chi_i^{[t+1,1]} = \text{Var}[\hat{p}^{[t+1,1]}(x_i)]$ ;
13  if  $\chi_i^{[t+1,1]} < h^{2[t+1,1]}$  then
14    Calculate  $\Lambda_i^{[t+1,1]}$  and  $\gamma_i^{[t+1,1]}$  using Eq. (12),
15  else
16    Calculate  $\Lambda_i^{[t+1,1]}$  and  $\gamma_i^{[t+1,1]}$  using Eq. (13).
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$$\gamma_i^{[t+1,1]} = \beta \left(\frac{\eta_i^{[t+1,1]}}{\chi_i^{[t+1,1]}} - \frac{t_i^{[t+1,1]}}{h_i^{2[t+1,1]}} \right) + (1 - \beta) \gamma_i^{[t,1]}, \quad (12b)$$

where β is the damping factor. On the other hand, if $\chi_i^{[t+1,1]} > h_i^{2[t+1,1]}$, MM would bring negative variance, which is illegal to be retained for next loop, and indeed indicates that the detector and decoder strongly disagree. In this case, following [5], we renew $\Lambda_i^{[t+1,1]}$ and $\gamma_i^{[t+1,1]}$ by a weighted Gaussian projection of $p_r^{[t+1]}(x_i)$ as

$$\Lambda_i^{[t+1,1]} = \varphi (1 / \text{Var}_{p_r^{[t+1]}}[(x_i)]), \quad (13a)$$

$$\gamma_i^{[t+1,1]} = \varphi (\mathbb{E}_{p_r^{[t+1]}}[(x_i)] / \text{Var}_{p_r^{[t+1]}}[(x_i)]), \quad (13b)$$

where $0 < \varphi < 1$ is an attenuation factor. With Eq. (13), the mean of the obtained $p^{[t+1]}(x_i)$ is indeed unchanged, which still matches $p_r^{[t+1]}(x_i)$. Meanwhile, the variance becomes $1/\Lambda_i^{[t+1,1]} = \text{Var}_{p_r^{[t+1]}}[(x_i)]/\varphi$, which is enlarged by a factor of $1/\varphi$. The proposed EPA IDD scheme is shown in Alg. 1.

B. The Modified BP Flip Decoder for IDD

In this study, we employ a non-resetting scheme [13] in the BP decoder to accelerate IDD convergence and reduce the required decoding iterations. When the outer loop of IDD reaches a certain number, it is observed that the LLR matrices $\mathbf{R}$ and $\mathbf{L}$ in the polar BP decoder tend to converge. In other words, the elements of $\mathbf{R}$ and $\mathbf{L}$ tend to be amplified without flip of signs. At this point, if the decoding still fails, the error is often unable to be corrected with the next few rounds of iteration. In this case, the a priori LLRs of specific information bits need to be changed to improve the error correction performance. For separate polar decoding, BPF progressively flips one or more information bits in CS and then performs conventional BP decoding. Here, a similar but simpler method,

as described in Alg. 2, is proposed for both EP- and EPA-based IDD without the exhausting enumeration operations in [9].

Firstly, a one-dimensional CS can be constructed offline as the input to the BP decoder. The CS mentioned here contains the first bit of each rate-1 node in the polar code [10]:

$$\text{CS} = \bigcup_j \mathcal{R}_j[1], \quad (14)$$

where $\mathcal{R}[1]$ denotes the rate-1 node and j denotes the index of the rate-1 node. Notably, the rate-1 node referred to in this paper is separated from the SPC node, Rep node, and rate-0 node, which largely reduces the total number of bits in CS denoted as $|\text{CS}|$. After the IDD outer loop reaches a certain number α , if the early termination criterion still fails, we traverse the elements of $L_{i^*,0}$ in the BP decoding, where $i^* \in \text{CS}$. With $L_{i^*,0} > 0$, the a priori LLR of the i^* -th bit is set to $-\infty$. With $L_{i^*,0} < 0$, the a priori LLR of the i^* -th bit is set to $+\infty$. i.e., we lose trust in the correctness of the values of $L_{i^*,0}$ and flip the corresponding information bits. Since the bits in CS are supposed to be error-prone, if an error occurs in CS, there is a high probability that some other information bits will be erroneously decoded during the decoding process. On the other hand, if the erroneous bits in CS are correctly flipped, it is very likely that some other incorrect non-CS information bits in the same sub-node will be corrected with the subsequent BP decoding iterations. Till the end of the IDD outer loops, if the decoding result still fails, the belief-enhancing operation is then performed, as we assume at this point the information bits in CS should be reliable while an error may arise among other bits. Specifically, with $L_{i^*,0} > 0$, the a priori LLR of the i^* -th bit is set to $+\infty$. With $L_{i^*,0} < 0$, the a priori LLR of the i^* -th bit is set to $-\infty$. The information bits are sequentially enhanced and then decoded until the decoder output satisfies the early termination criterion or the maximum of decoding iterations is reached.

IV. SIMULATION RESULTS

This section presents the bit-error-rate (BER) performance and complexity of the proposed EPA-IDD with mBPF (EPA-mBPF) and DEP with mBPF (DEP-mBPF), compared with regular DEP-BP, MMSE-PIC with BP, and SDD schemes EP-SCL8 (SCL with list size of 8) and EP-BP. Here we consider the correlated channel with $\zeta_R = 0.2$ and $\zeta_T = 0.05$. The polar codes are encoded with length $N = 1024$ and rate $R = 900/1024$, and concatenated with cyclic redundancy check (CRC) as early termination criterion with length of 11.

A. Numerical Results

Fig. 1 shows the comparisons of BER performance under $N_t \times N_r = 32 \times 64$, 64×112 , and 64×128 scenarios with 256-QAM. The iteration number of the outer loop T_{max} is 10 with $S = 1$ and $I_{max} = 3$. In Figs 1(a) and 1(c), DEP-mBPF outperforms DEP-BP for 0.2 dB and EP-SCL8 for 0.4 dB at the BER of 1×10^{-4} . In Fig. 1(b), DEP-mBPF outperforms DEP-BP for 0.1 dB and EP-SCL8 for 0.4 dB at the BER of 1×10^{-3} , and the gap between them becomes larger with the

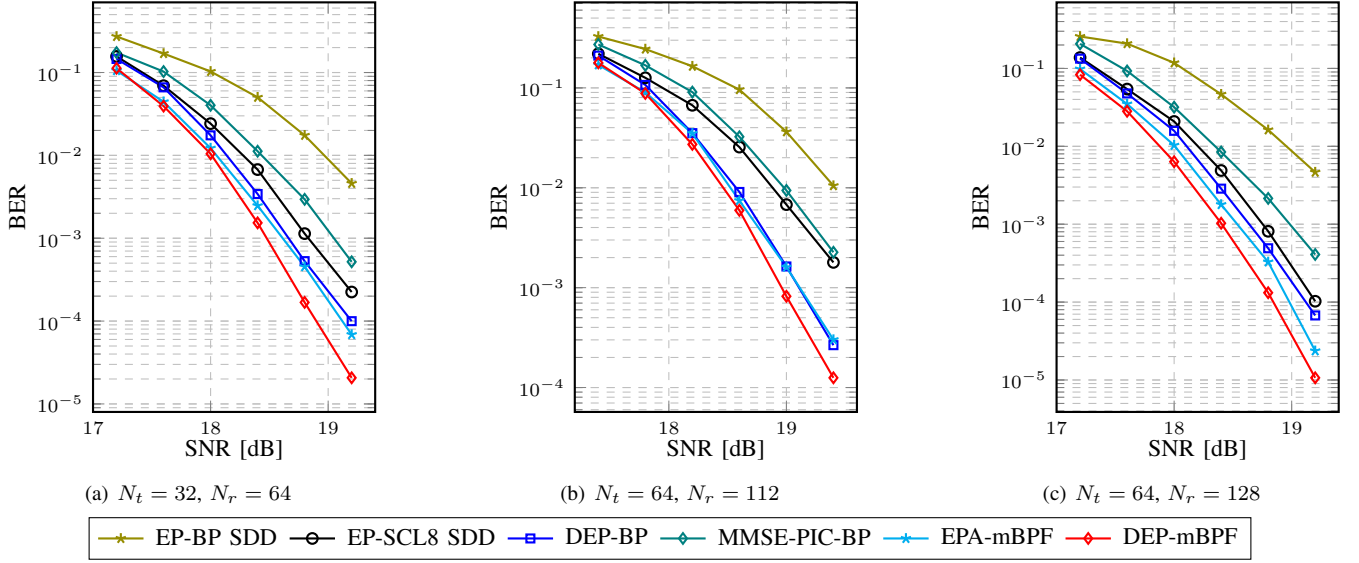


Fig. 1. BER performance comparison of EP-SCL8, EP-BP SDD, DEP-BP, MMSE-PIC-BP, and the proposed EPA-mBPF and DEP-mBPF, under 32×64 , 64×112 , and 64×128 polar-coded MIMO with 256-QAM, $N = 1024$, and $R = 900/1024$.

Algorithm 2: The proposed mBPF decoder

Input : $\text{LLR}, \mathcal{A}^c, \text{CS}, I_{max}, t, \mathbf{R}, \mathbf{L}$.
Output: $\hat{\mathbf{a}}, \mathbf{R}, \mathbf{L}$.

- 1 Initialize the BP decoder.
- 2 $\hat{\mathbf{a}}, \mathbf{R}, \mathbf{L} \leftarrow \text{conventional-BP-decoder}(\text{LLR}, \mathcal{A}^c, I_{max}, \mathbf{R}, \mathbf{L})$.
- 3 **if** $\hat{\mathbf{a}}$ satisfies CRC **then**
- 4 **break**;
- 5 **else**
- 6 **if** $t > \alpha$ **then**
- 7 **for** $i = 0, 1, \dots, |\text{CS}| - 1$ **do**
- 8 $\mathbf{R}_{\text{CS},0} = 0$;
- 9 $\mathbf{R}_{\text{CS}(i),0} = -\text{sgn}(\mathbf{L}_{\text{CS}(i),0}) \cdot \text{realmax}$;
- 10 $\hat{\mathbf{a}}, \mathbf{R}, \mathbf{L} \leftarrow \text{conventional-BP-decoder}(\text{LLR}, \mathcal{A}^c, I_{max}, \mathbf{R}, \mathbf{L})$;
- 11 **if** $\hat{\mathbf{a}}$ satisfy CRC **then**
- 12 **break**;
- 13 **if** $\hat{\mathbf{a}}$ does not satisfy CRC and $t = T_{max}$ **then**
- 14 $\mathbf{R}_{:,1:\text{end}} = 0$; $\mathbf{L}_{:,1:\text{end}-1} = 0$;
- 15 **for** $i = 0, 1, \dots, |\text{CS}| - 1$ **do**
- 16 $\mathbf{R}_{\text{CS},0} = 0$; $\mathbf{R}_{\text{CS}(i),0} = \text{sgn}(\mathbf{L}_{\text{CS}(i),0}) \cdot \text{realmax}$;
- 17 $\hat{\mathbf{a}}, \mathbf{R}, \mathbf{L} \leftarrow \text{conventional-BP-decoder}(\text{LLR}, \mathcal{A}^c, I_{max}, \mathbf{R}, \mathbf{L})$;
- 18 **if** $\hat{\mathbf{a}}$ satisfy CRC **then**
- 19 **break**;

increase of SNR. Moreover, the performances of EPA-mBPF and DEP-BP are relatively close, both attaining about 0.2 dB gain compared with EP-SCL8, while EPA-mBPF outperforms DEP-BP as the SNR increases. In conclusion, these results confirm that EPA-mBPF attains robust good BER performance with lower complexity, and DEP-mBPF can achieve the best performance of all at the expense of detection complexity.

B. Complexity Analysis

According to the complexity of EP and EPA analyzed in [4], with $S = 1$, in each outer loop, DEP would cost one extra inversion of $\mathcal{O}(N_t^3)$ than EPA in the detection module. Fig. 2 illustrates the decoding complexity comparisons among EPA-mBPF, DEP-mBPF, DEP-BP, and EP-BP SDD. Under the same SNR, the required averaged BP decoding iterations per codeword to achieve four different BERs are compared. As shown in Fig. 2, EP-BP SDD requires the most decoding iterations when $\text{BER} = 7 \times 10^{-3}$. However, when BER is less than 10^{-3} , EP-BP SDD can no longer achieve the performance even with more iterations. With $\text{BER} = 7 \times 10^{-4}$ and 3×10^{-4} , DEP-mBPF and DEP-BP share the same number of averaged iterations while EPA-mBPF needs a few more. With regard to lower $\text{BER} = 1.5 \times 10^{-4}$, the decoding iterations needed by DEP-mBPF is fewer than that of DEP-BP. Meanwhile, EPA-mBPF can achieve the same BER performance as DEP-BP with just a negligible increase of decoding iterations but considerable decrease of detection complexity. Moreover, DEP-mBPF requires only about 12 decoding iterations on average to achieve the best performance with $\text{BER} = 1 \times 10^{-4}$, which other schemes cannot attain. The discussion above proves the high efficiency of the proposed algorithms.

V. HARDWARE ARCHITECTURE

To analyze the implementation complexity of the proposed IDD receiver, we propose an efficient hardware architecture in this section. Fig. 3 shows the overall architecture of EPA-mBPF, which can be divided into three parts: 1) MIMO EPA detector, 2) polar BP decoder, and 3) joint detection and decoding. Furthermore, the intermediate messages of the detector and decoder for each iteration are stored in a joint memory that includes both decoding and detection memories.

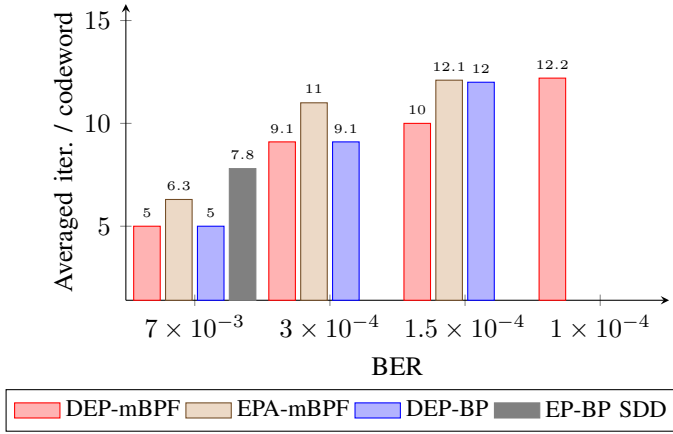


Fig. 2. Decoding iteration comparison at the SNR of 21.8 dB under 64×112 and 256-QAM. Notice that some algorithms are not presented at certain BERs since the specified performance cannot be achieved.

The EPA detector consists of preprocessing units, NSA units that approximate the initial matrix inversion with Neumann Series, t -initialization units, and EPA units. The preprocessing units use systolic arrays to perform matrix multiplication, and then send the preprocessing messages $\mathbf{A}$, $\tilde{\mathbf{b}}$, $\mathbf{v}$ to the NSA units to compute the mean $\boldsymbol{\mu}$ and variance $\boldsymbol{\Sigma}$. Based on the aforementioned messages, the EPA unit provides the symbol soft messages required for polar BP decoding.

The polar BP decoder receives the symbol soft message from the EPA detector and applies a symbol-to-LLR message transform to perform BP decoding. During the iterative decoding process, non-resetting decoding messages are stored in the L/R memory. The prior message R_0 can be modified by the CS using our proposed flipping scheme. After BP decoding, the LLR soft messages are subjected to the corresponding inverse transform, which are fed to the joint detection and decoding for symbol probability density computations.

The joint detection and decoding module provide two symbol probability densities for the update of Λ and γ . Gaussian projection and MM are implemented in parallel to output two kinds of Λ and γ calculated by Eq.s (12) and (13), respectively, before being selected by the multiplexer array.

VI. CONCLUSION

In this paper, an efficient IDD receiver for polar-coded MIMO systems is proposed with its corresponding hardware architecture, which shows enhanced BER performance with lower computational complexity. In future work, we will focus on constructing adaptive CSs for different outer loops to further improve the IDD performance.

ACKNOWLEDGMENT

This work was supported in part by National Key R&D Program of China under Grant 2020YFB2205503, in part by NSFC under Grants 62331009, 62122020, and 62001108, in part by the Jiangsu Provincial NSF under Grant BK20211512,

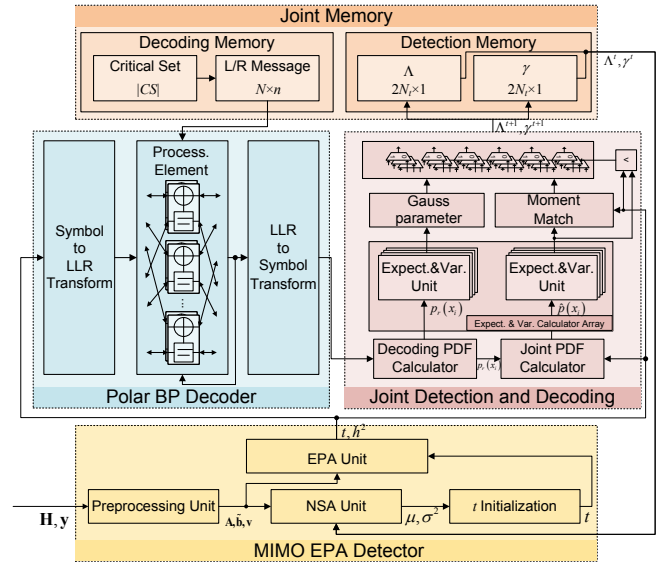


Fig. 3. Overall architecture of the EPA-mBPF IDD receiver.

in part by the Major Key Project of PCL under Grant PCL2021A01-2, and in part by the Fundamental Research Funds for the Central Universities.

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